

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Mock Interview

**COURSE NAME: Cloud Computing and
Big Data Analytics**

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO5: Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.

(BL5)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Articulation (Level 4)

Question Count: 10

Total Marks: 25

Code of Conduct

I S. Nitheesh(192524329) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student S. Nitheesh

1. Explain the difference between HDFS and traditional file systems.

Answer:

HDFS (Hadoop Distributed File System) is designed to store very large amounts of data across multiple machines, whereas traditional file systems usually store data on a single computer or storage device.

Key differences:

- HDFS distributes files across multiple **DataNodes**.
- HDFS uses **replication** for fault tolerance.
- It is designed for **large files and sequential access**.
- Traditional file systems generally depend on a single server or storage system.
- HDFS can scale horizontally by adding more DataNodes.

Architecture:

Client → NameNode → Multiple DataNodes

2. Why is replication important in distributed storage systems?**Answer:**

Replication means keeping multiple copies of the same data on different nodes. It is important because individual machines can fail.

For example, with a **replication factor of 3**, three copies of a block are stored on different DataNodes.

Benefits:

- Provides **fault tolerance**.
- Improves **data availability**.
- Prevents data loss when a node fails.
- Can improve read performance because data can be read from different replicas.

Architecture:

Data Block → Node 1 + Node 2 + Node 3

3. How does MapReduce divide work between map and reduce phases?**Answer:**

MapReduce divides a large data-processing task into two main phases.

Map phase:

- Input data is divided into smaller blocks.

- Multiple Mapper tasks process these blocks in parallel.
- Each Mapper generates key-value pairs.

Example:

Input logs → Mapper → (Error, 1)

Reduce phase:

- The Shuffle and Sort stage groups similar keys.
- Reducers process the grouped values.

Example:

(Error, [1,1,1,1]) → Reducer → (Error, 4)

Architecture:

Input → Map → Shuffle & Sort → Reduce → Output

This parallel design allows Hadoop to process very large datasets efficiently.

4. What role does YARN play in large-scale Hadoop processing?

Answer:

YARN stands for **Yet Another Resource Negotiator**. It manages cluster resources and schedules applications in a Hadoop cluster.

Its main responsibilities are:

- Allocating **CPU and memory**.
- Scheduling applications.
- Managing application execution.
- Monitoring running applications.
- Handling failures and restarting tasks when necessary.

Main components:

- **ResourceManager:** Manages overall cluster resources.
- **NodeManager:** Manages resources on individual nodes.
- **ApplicationMaster:** Manages a particular application.

Architecture:

Client → ResourceManager → NodeManagers → Application Containers

5. Differentiate NAS and SAN in an enterprise data environment.

Answer:

NAS

Network Attached Storage

Provides file-level storage

Accessed using file protocols

Easier to manage

Common for shared files

Usually uses Ethernet networks

SAN

Storage Area Network

Provides block-level storage

Accessed using storage protocols

More complex

Common for databases and enterprise applications

Often uses dedicated high-speed storage networks

Simple example:

NAS is like a **shared network folder**, while SAN provides storage that appears to servers more like **local disks**.

6. What is the purpose of ETL in a big data pipeline?

Answer:

ETL means **Extract, Transform, Load**.

- **Extract:** Collect data from different sources.
- **Transform:** Clean, validate, filter, and convert the data.
- **Load:** Store the processed data in a target system such as HDFS or a data warehouse.

Example:

Database + CSV + API → Extract → Clean/Transform → HDFS/Data Warehouse

ETL ensures that data is **clean, consistent, and ready for analysis**.

7. How does Apache NiFi help in data integration workflows?

Answer:

Apache NiFi is a data-flow and integration platform used to move and process data between different systems.

It provides:

- Data ingestion from databases, files, APIs, and applications.
- Data routing and transformation.
- Monitoring of data flows.
- Error handling and retry mechanisms.

- Back-pressure to control data flow.
- Data provenance to track where data came from.

Architecture:

Source → NiFi → Validate/Transform → HDFS/Database/Cloud

NiFi is particularly useful when an organization has **many different data sources and destinations**.

8. What are the key failure points in a Hadoop cluster and how would you handle them?

Answer:

The major failure points include **DataNodes, NameNodes, NodeManagers, disks, and network connections**.

Handling failures:

1. **DataNode failure:**
HDFS automatically creates new replicas from healthy copies.
2. **NameNode failure:**
Use **NameNode High Availability** with Active and Standby NameNodes.
3. **Disk failure:**
Use multiple disks and HDFS replication.
4. **NodeManager failure:**
YARN detects the failed node and reschedules tasks on healthy nodes.
5. **Network failure:**
Use redundant network connections and proper rack-aware architecture.

Overall approach:

Failure Detection → Recovery → Replication/Rescheduling → Monitoring

9. How would you design a secure data ingestion pipeline for a large organization?

Answer:

I would design the pipeline with security at every stage.

Architecture:

Data Sources → Secure NiFi → Validation → HDFS → Processing → Analytics

Security measures:

- Use **TLS/SSL** for data in transit.
- Use **Kerberos** for authentication in Hadoop.

- Apply role-based access control.
- Encrypt sensitive data at rest.
- Validate and sanitize incoming data.
- Maintain audit logs and data provenance.
- Use network segmentation and firewalls.
- Implement backup and disaster recovery.

Example:

A database sends data through an encrypted connection to NiFi. NiFi validates and routes it to HDFS, where only authorized users can access the data.

10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Answer:

I would explain it using a simple example:

"Imagine we have a huge warehouse containing millions of boxes. Instead of asking one person to handle every box, we divide the boxes among many workers. Each worker processes a portion of the boxes at the same time. If one worker is unavailable, another can handle the work."

In Hadoop:

- **HDFS** is the warehouse where data is stored across many machines.
- **YARN** is the manager that distributes computing resources.
- **MapReduce/Spark** are the workers that process the data.
- **Replication** keeps backup copies of important data.
- Adding more machines allows the system to handle more data.